

Two-dimensional Random Variables

Definition: Let X and Y be two r.v. defined on the same sample space S , then the function (X, Y) that assigns a point in $R^2 (= R \times R)$, is called a two-dimensional random variable.

When (X, Y) is a two-dimensional discrete r.v., the possible values of (X, Y) may be represented as (x_i, y_j) , $i = 1, 2, \dots, n$; $j = 1, 2, \dots, m$.

If (X, Y) can assume all values in a specified region R in the xy –plane, (X, Y) is called a two-dimensional continuous R.V.

Joint Probability Function

Let X and Y be random variables on a sample space S with respective image sets $X(S) = \{x_1, x_2, \dots, x_n\}$ and $Y(S) = \{y_1, y_2, \dots, y_m\}$.

$$X(S) \times Y(S) = \{x_1, x_2, \dots, x_n\} \times \{y_1, y_2, \dots, y_m\}$$

Into a probability space by defining the probability of the ordered pair to be $\mathbf{P}(X = \mathbf{x}_i, Y = \mathbf{y}_j)$ or $\mathbf{p}(\mathbf{x}_i, \mathbf{y}_j)$.

The function $\mathbf{p}$ on $X(S) \times Y(S)$ is defined by:

$p_{ij} = P(X = x_i \cap Y = y_j) = p(x_i, y_j)$ is called the
joint probability function of $\mathbf{X}$ and $\mathbf{Y}$.

$X \backslash Y$	y_1	y_2	y_3	$\dots$	y_j	$\dots$	y_m	Total
x_1	p_{11}	p_{12}	p_{13}	$\dots$	p_{1j}	$\dots$	p_{1m}	$p_{1\cdot}$
x_2	p_{21}	p_{22}	p_{23}	$\dots$	p_{2j}	$\dots$	p_{2m}	$p_{2\cdot}$
x_3	p_{31}	p_{32}	p_{33}	$\dots$	p_{3j}	$\dots$	p_{3m}	$p_{3\cdot}$
$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\dots$	$\cdot$	$\dots$	$\cdot$	$\vdots$
$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\dots$	$\cdot$	$\dots$	$\cdot$	$\vdots$
$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\dots$	$\cdot$	$\dots$	$\cdot$	$\vdots$
x_i	p_{i1}	p_{i2}	p_{i3}	$\dots$	p_{ij}	$\dots$	p_{im}	$p_{i\cdot}$
$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\dots$	$\cdot$	$\dots$	$\cdot$	$\vdots$
$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\dots$	$\cdot$	$\dots$	$\cdot$	$\vdots$
$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\dots$	$\cdot$	$\dots$	$\cdot$	$\vdots$
x_n	p_{n1}	p_{n2}	p_{n3}	$\dots$	p_{nj}	$\dots$	p_{nm}	$p_{n\cdot}$
Total	$p_{\cdot 1}$	$p_{\cdot 2}$	$p_{\cdot 3}$	$\dots$	$p_{\cdot j}$	$\dots$	$p_{\cdot m}$	1

Definition:

If (X, Y) is a two-dimensional discrete random variable, then the joint discrete function of X, Y also called the joint probability mass function of (X, Y) denoted by p_{XY} is defined as:

$$p_{XY}(x_i, y_j) = P(X = x_i, Y = y_j) \text{ for a value of } (x_i, y_j) \text{ of } (X, Y)$$

$$p_{XY}(x_i, y_j) = 0, \text{ otherwise.}$$

$$\sum \sum p_{XY}(x_i, y_j) = 1, \text{ where the summation is taken over all possible values of } (X, Y).$$

Marginal Probability Function:

$$\begin{aligned} p_X(x_i) &= P(X = x_i) \\ &= P(X = x_i \cap Y = y_1) + P(X = x_i \cap Y = y_2) + \dots + \\ &\quad P(X = x_i \cap Y = y_m) \end{aligned}$$

$$p_X(x_i) = p_{i1} + p_{i2} + \dots + p_{ij} + \dots + p_{im} = \sum_{j=1}^m p_{ij} = p_{i.}$$

Also,
$$\sum_{i=1}^n p_{i.} = p_{1.} + p_{2.} + \dots + \dots + p_{n.} = \sum_{i=1}^n \sum_{j=1}^m p_{ij} = 1$$

Similarly,
$$p_Y(y_j) = P(Y = y_j) = \sum_{i=1}^n p_{ij} = p_{.j}$$

which is the marginal probability function of Y.

Conditional Probability Function:

The conditional probability mass function of X , given $Y = y$ is defined as:

$$p_{X|Y}(x|y) = \frac{P(X=x, Y=y)}{P(Y=y)}, \text{ provided } P(Y=y) \neq 0$$

Similarly,
$$p_{Y|X}(y|x) = \frac{P(X=x, Y=y)}{P(X=x)}$$

A necessary and sufficient condition for the discrete random variables X and Y to be independent is:

$$P(X = x_i, Y = y_j) = P(X = x_i) P(Y = y_j) \text{ for all values } (x_i, y_j) \text{ of } (X, Y).$$

Joint Probability Density Function

If (X, Y) is two-dimensional continuous RV such that,

$$P\left(x - \frac{dx}{2} \leq X \leq x + \frac{dx}{2} \text{ and } y - \frac{dy}{2} \leq Y \leq y + \frac{dy}{2}\right) = f(x, y)dx dy$$

Then $f(x, y)$ is called the joint pdf of (X, Y) , provided $f(x, y)$ satisfies the following conditions:

(i) $f(x, y) \geq 0$, for all $(x, y) \in R$, where R is the range space.

(ii) $\iint_R f(x, y) dx dy = 1$.

Two-dimensional distribution function

Definition: The distribution function of the two-dimensional random variable (X,Y) is a real valued function F defined for all real x and y by the relation:

$$F_{XY}(x, y) = P(X \leq x, Y \leq y) = \sum_{y_j \leq y} \sum_{x_i \leq x} p_{ij}$$

In the continuous case,

$$F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(x, y) dx dy$$

Properties: $\frac{\partial^2 F}{\partial x \partial y} = f(x, y)$

Marginal Distribution Functions:

$$F_X(x) = P(X \leq x) = P(X \leq x, Y < \infty) = \lim_{y \rightarrow \infty} F_{XY}(x, y) = F_{XY}(x, \infty)$$

$$\text{Similarly, } F_Y(y) = P(Y \leq y) = P(X < \infty, Y \leq y) = \lim_{x \rightarrow \infty} F_{XY}(x, y) = F_{XY}(\infty, y)$$

In case of joint discrete r.v., the marginal distribution functions are:

$$F_X(x) = \sum_y P(X \leq x, Y = y) \text{ and } F_Y(y) = \sum_x P(X = x, Y \leq y)$$

In case of Jointly continuous r.v.,

$$F_X(x) = \int_{-\infty}^x \left\{ \int_{-\infty}^{\infty} f_{XY}(x, y) dy \right\} dx \quad F_Y(y) = \int_{-\infty}^y \left\{ \int_{-\infty}^{\infty} f_{XY}(x, y) dx \right\} dy$$

Marginal Density function:

$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$ - is the marginal density function of X.

$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$ - is the marginal density function of Y

$$\begin{aligned} P(a \leq X \leq b) &= P(a \leq X \leq b, -\infty \leq Y \leq \infty) \\ &= \int_a^b \left\{ \int_{-\infty}^{\infty} f_{XY}(x, y) dy \right\} dx \\ &= \int_a^b f_X(x) dx \end{aligned}$$

$$P(a \leq X \leq b) = \int_c^d f_Y(y) dy$$

Independent RVs

If (X, Y) is two-dimensional independent RVs,

$$P(X = x, Y = y) = P(X = x) \cdot P(Y = y) \quad \text{- Discrete}$$

$$f(x, y) = f(x) \cdot f(y) \quad \text{- Continuous}$$

Example 10: For a bivariate probability distribution of (X, Y) given below,

X \ Y	1	2	3	4	5	6
0	0	0	1/32	2/32	2/32	3/32
1	1/16	1/16	1/8	1/8	1/8	1/8
2	1/32	1/32	1/64	1/64	0	2/64

Find $P(X \leq 1)$

$P(Y \leq 3)$

$P(X \leq 1, Y \leq 3)$

$P(X \leq 1 | Y \leq 3)$

$P(Y \leq 3 | X \leq 1)$

$P(X + Y \leq 4)$

$$\begin{aligned}
 P(X \leq 1) &= P(X = 0) + P(X = 1) \\
 &= \sum_{j=1}^6 P(X = 0, Y = j) + \sum_{j=1}^6 P(X = 1, Y = j) = \frac{1}{4} + \frac{5}{8} = \frac{7}{8}
 \end{aligned}$$

$$\begin{aligned}
 P(Y \leq 3) &= P(Y = 1) + P(Y = 2) + P(Y = 3) \\
 &= \sum_{i=0}^2 P(X = i, Y = 1) + \sum_{i=0}^2 P(X = i, Y = 2) + \sum_{i=0}^2 P(X = i, Y = 3) \\
 &= \frac{3}{32} + \frac{3}{32} + \frac{11}{64} = \frac{23}{64}
 \end{aligned}$$

$$\begin{aligned}
 P(X \leq 1, Y \leq 3) &= \sum_{j=1}^3 P(X = 0, Y = j) + \sum_{j=1}^3 P(X = 1, Y = j) \\
 &= \frac{1}{32} + \frac{1}{4} = \frac{9}{32}
 \end{aligned}$$

$$P(X \leq 1 | Y \leq 3) = \frac{P(X \leq 1, Y \leq 3)}{P(Y \leq 3)} = \frac{9/32}{23/64} = \frac{18}{23}$$

$$P(Y \leq 3 | X \leq 1) = \frac{P(X \leq 1, Y \leq 3)}{P(X \leq 1)} = \frac{9/32}{7/8} = \frac{9}{28}$$

$$\begin{aligned} P(X + Y \leq 4) &= \sum_{j=1}^4 P(X = 0, Y = j) + \sum_{j=1}^3 P(X = 1, Y = j) + \sum_{j=1}^2 P(X = 2, Y = j) \\ &= \frac{3}{32} + \frac{1}{4} + \frac{1}{16} = \frac{13}{32} \end{aligned}$$

Example 11:

Three balls are drawn at random without replacement from a box containing 2 white, 3 red and 4 black balls. If X denotes the number of white balls drawn and Y denotes the number of red balls drawn. Find the joint probability distribution of (X, Y) .

Solution:

$$\begin{aligned}P(X = 0, Y = 0) &= P(\text{drawing 3 balls none of which is white or red}) \\&= P(\text{all the 3 balls are black}) \\&= \frac{{}^4C_3}{{}^9C_3} = \frac{1}{21}\end{aligned}$$

$$P(X = 0, Y = 1) = P(\text{drawing 1 red and 2 black balls}) = \frac{3}{14}$$

X \ Y	0	1	2	3
0	$\frac{1}{21}$	$\frac{3}{14}$	$\frac{1}{7}$	$\frac{1}{84}$
1	$\frac{1}{7}$	$\frac{2}{7}$	$\frac{1}{14}$	0
2	$\frac{1}{21}$	$\frac{1}{28}$	0	0

Example 13

If X and Y are two random variables having joint density function:

$$f(x, y) = \begin{cases} \frac{1}{8}(6 - x - y) & ; 0 \leq x \leq 2, 2 \leq y \leq 4 \\ 0 & \end{cases}$$

Find the (i) Marginal density functions of X and Y

(ii) $P(X < 1 \cap Y < 3)$

(iii) $P(X + Y < 3)$ (iv) $P(X < 1 \mid Y < 3)$

Example 13

The joint pdf of a two-dimensional RV (X, Y) is given by

$$f(x, y) = xy^2 + \frac{x^2}{8} \quad ; \quad 0 \leq x \leq 2, \quad 0 \leq y \leq 1$$

Compute $P(X > 1)$, $P(Y < \frac{1}{2})$, $P(X > 1 | Y < \frac{1}{2})$, $P(X < Y)$ and $P(X + Y \leq 1)$

Solution:

$$P(X > 1) = \frac{19}{24}$$

$$P\left(Y < \frac{1}{2}\right) = \frac{1}{4}$$

$$P(X > 1, Y < \frac{1}{2}) = \frac{5}{24}$$

$$P(X > 1 | Y < \frac{1}{2}) = \frac{5/24}{1/4} = \frac{5}{6}$$

$$P(X < Y) = \frac{53}{480}$$

$$P(X + Y \leq 1) = \frac{13}{480}$$

Example 14

Joint density function of X and Y is given by:

$$f(x, y) = 4xye^{-(x^2+y^2)} \quad ; \quad x \geq 0, y \geq 0$$

Test whether X and Y are independent.

Find the conditional density function of X given $Y = y$.